

Algorithm: Random Forest, Decision Tree (Hyperparameter Tuning)

1. Business Understanding

Objective

The target of this study is to perform Random Forest algorithm and compare its working performance with Hyperparameter Tuning, Feature Importance. Such comparison also enables the user to gain insights into the internal working of ensemble methods, especially Random Forest.

2. Data Understanding

Dataset: [Germany Used Cars Dataset 2023]

(<https://www.kaggle.com/datasets/wspirat/germany-used-cars-dataset-2023>)

Dataset: Collection of car offers from one of Germany's largest car sales websites, AutoScout24. This scraped dataset contains a wide range of information about car offers, covering a cars manufactured from 1995 to 2023.

3. Data Preparation

This step follows:

- A. Handle missing values.
- B. Encode categorical features.
- C. Remove outliers by IQR technique
- D. Scale numerical features using **stdscaler**
- E. Split into train & test sets.

4. Modeling - Baseline Implementation

Decision Tree focuses on splitting data based on Gini impurity to make predictions, while a Random Forest combines multiple decision trees for improved accuracy and robustness through majority voting

5. Evaluation

Without hyperparameter tuning:

Random Forest (scikit-learn) Accuracy: 0.9090

	precision	recall	f1-score	support
0	0.97	0.96	0.97	1344
1	0.58	0.64	0.61	157
2	0.71	0.68	0.69	121
3	0.00	0.00	0.00	4
accuracy			0.91	1626
macro avg	0.57	0.57	0.57	1626
weighted avg	0.91	0.91	0.91	1626

With hyperparameter tuning,

Fitting 3 folds for each of 108 candidates, totaling 324 fits

Best Hyperparameters:

{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}

Random Forest (scikit-learn - Tuned) Accuracy: 0.9090

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Feature importance of this dataset:

Feature Importances: [0.08855558 0.13076937 0.0357671 0.07289044 0.07608824 0.14598291 0.1588236 0.01160966 0.06142918 0.05046902 0.0760674 0.09154748]		
	Feature	Importance
6	power_ps	0.158824
5	power_kw	0.145983
1	model	0.130769
12	offer_description	0.091547
0	brand	0.088556
4	year	0.076088
11	mileage_in_km	0.076067
3	registration_date	0.072890
9	fuel_consumption_l_100km	0.061429
10	fuel_consumption_g_km	0.050469
2	color	0.035767
8	fuel_type	0.011610
7	transmission_type	0.000000

6. Conclusion and future work

Random Forest is performing well. So, changing grid search values furthermore, we can achieve better robust model (but it takes more time to train model which computationally expensive for small datasets).